Worst-Group Conformal Reliability under Spurious Correlation Is a Calibration Problem, Not a Representation Problem: A Confound-Controlled Multi-Backbone Study

One Octadion[1*], Novanto Yudistira[1] and Yuta Nakashima[2]

[1*]Universitas Brawijaya, Malang, Indonesia.
[2]SANKEN, The University of Osaka, Osaka, Japan.

*Corresponding author(s). E-mail(s): octadion@student.ub.ac.id;
Contributing authors: yudistira@ub.ac.id;
n-yuta@im.sanken.osaka-u.ac.jp;

Abstract

Group-robust training methods are certified almost exclusively by worst-group *accuracy*, while a parallel line of conformal-prediction theory shows that post-hoc calibration can only *relocate*, never remove, cross-group heterogeneity in conformity scores. This invites a natural but untested intuition: that representations made robust on the accuracy side should *also* carry the conformal (coverage and set-size) reliability of the worst group. We test this intuition with a deliberately confound-controlled protocol across two benchmarks (Waterbirds, CelebA), two frozen backbones (an ERM-trained ResNet-50 and CLIP ViT-B/32), five last-layer training interventions (ERM, DFR, AFR, balanced subsampling, last-layer GroupDRO), three nonconformity scores (APS, RAPS, THR), three calibration policies (marginal split, Mondrian group-conditional, shift-robust), and a calibration-to-test correlation-strength sweep. Our central result is a clean dissociation that holds across every cell of this grid—both datasets, both backbones. *Worst-group coverage is a property of the calibration mechanism, not of the representation*: under marginal calibration, worst-group coverage swings by up to **0.36** across training methods (ERM collapses to **0.51**); under Mondrian calibration it is essentially flat across all training methods in every cell (cross-training spread $\leq$ **0.024**), and the *same* ERM model is lifted from **0.51**–**0.82** to **0.85**–**0.89** purely by changing the calibration policy. Training, in turn, buys *efficiency*, not coverage: under Mondrian the worst-group set size tracks base accuracy in three of four cells (correlation **−0.85** to **−1.00**). The remaining findings reinforce the dissociation: robust training's apparent reduction of cross-group conformity-score divergence is largely an accuracy effect that, once matched, is modest or unmeasurable; the accuracy-best training method is almost always the burden-best (the inversion is rare and marginal); under correlation-strength shift worst-group coverage is stable but persistently sub-target; and the spurious attribute remains almost perfectly recoverable from the deployed representation (**AUROC** $\in$ **[0.95, 0.99]**), so there is no coverage–auditability tension and group-conditional calibration is deployable with *predicted* groups (within **0.02** of the ground-truth oracle for robust models), needing no observed group labels at test time. The practical message is corrective and actionable: report conformal-burden comparisons at matched accuracy (uncontrolled comparisons overstate the training-side benefit by an order of magnitude), and rely on group-conditional calibration—which delivers worst-group coverage from ordinary, fully auditable representations.

Keywords: conformal prediction, spurious correlation, group robustness, worst-group coverage, group-conditional calibration

1 Introduction

Conformal prediction (Vovk et al. 2005; Angelopoulos et al. 2021; Romano et al. 2020) turns a point predictor into set-valued predictions with a distribution-free marginal coverage guarantee under exchangeability. Under a *spurious correlation*—a background, demographic, or context feature that co-varies with the label in training but not at deployment—marginal guarantees mask severe *worst-group* unreliability: the minority group whose spurious feature is mismatched is systematically under-covered or receives uninformatively large sets (Cresswell et al. 2025). Two largely separate programs respond. The first improves *training* so the worst group is accurate: GroupDRO (Sagawa* et al. 2020), Deep Feature Reweighting (DFR) (Kirichenko et al. 2023), AFR (automatic feature reweighting) (Qiu et al. 2023), JTT (Liu et al. 2021), CNC (Zhang et al. 2022), evaluated almost entirely on worst-group accuracy and calibration error (Yang et al. 2023). The second improves *calibration*: Mondrian and group-conditional conformal prediction (Vovk et al. 2005; Jung et al. 2023; Gibbs et al. 2024), Kandinsky calibration for overlapping and fractional groups (Bairaktari et al. 2025), multi-distribution robust conformal prediction (Yang and Jin 2026), adaptive fairness-aware conformal prediction (Zhou and Sesia 2024). A recent impossibility result formalizes the link: the calibration policy determines only *whether* cross-group score heterogeneity surfaces as coverage disparity or as set-size disparity, never whether it appears at all (Gao et al. 2026).

These two programs meet at a widespread but rarely tested intuition: if the worst group can be made *accurate* by changing the representation, perhaps the same change makes it *conformally reliable*, so robust training should pay off for downstream uncertainty quantification. A confound-controlled grid instead returns a clean *dissociation* (Figure 1)—the more useful result for a literature that tends to credit training-side robustness with conformal benefits it does not separately provide.

Our contributions are:

1. **Coverage is a calibration property, demonstrated (Section 6.1).** Holding the model and representation fixed and varying only the calibration policy, worst-group coverage swings by up to 0.36 across training methods under marginal calibration but collapses to a spread of $\leq$ 0.024 under Mondrian in *every* dataset$\times$backbone cell; the same ERM model is lifted by +0.06 to +0.35 purely by switching to group-conditional calibration. The representation does not deliver worst-group coverage; the calibration mechanism does.
2. **Training buys efficiency, not coverage (Section 6.2).** Under Mondrian (coverage matched across methods), the worst-group set size tracks base accuracy in three of four cells (corr $\in [-1.00, -0.85]$); training/accuracy improves *efficiency*, not coverage.
3. **A confound-controlled protocol, and "transfer" as an accuracy effect (Section 6.3).** We report a cross-group conformity-score divergence *matched on base accuracy*; uncontrolled comparisons overstate the conformal benefit of robust training by an order of magnitude, and once matched the residual is modest (Waterbirds) or structurally undefined (CelebA, where robust arms raise accuracy beyond ERM).
4. **Accuracy is a good proxy for conformal burden (Section 6.4).** The ranking inversion between worst-group accuracy and worst-group conformal burden appears in only 2 of 4 cells and is marginal where it appears.
5. **Stable but sub-target under shift; auditable and deployable (Sections 6.5–6.7).** Under correlation-strength shift, worst-group coverage is stable yet persistently sub-target; the spurious attribute is almost perfectly recoverable from the deployed representation (AUROC $\in [0.95, 0.99]$), so worst-group coverage and post-hoc group auditing coexist; and because the group is recoverable, group-conditional calibration is deployable with probe-predicted groups—worst-group coverage within 0.02 of the ground-truth-group oracle for 16 of 18 method$\times$cell combinations—so it does not require observed group labels at test time.

The unifying thesis (Section 7) is that, under spurious correlation, worst-group conformal reliability is governed by the calibration *mechanism*, not by the score *representation*. We frame the paper as a characterization rather than a method.

2 Related Work

Group-robust training.

Worst-group accuracy is improved by distributionally robust optimization (Sagawa* et al. 2020), last-layer retraining on group-balanced data (DFR) (Kirichenko et al. 2023), automatic feature reweighting (AFR) (Qiu et al. 2023), and pseudo-group methods such as JTT (Liu et al. 2021) and CNC (Zhang et al. 2022). The standard subpopulation-shift benchmark (Yang et al. 2023) compares these on accuracy-family metrics and calibration error but reports no set-valued or coverage quantities. Our

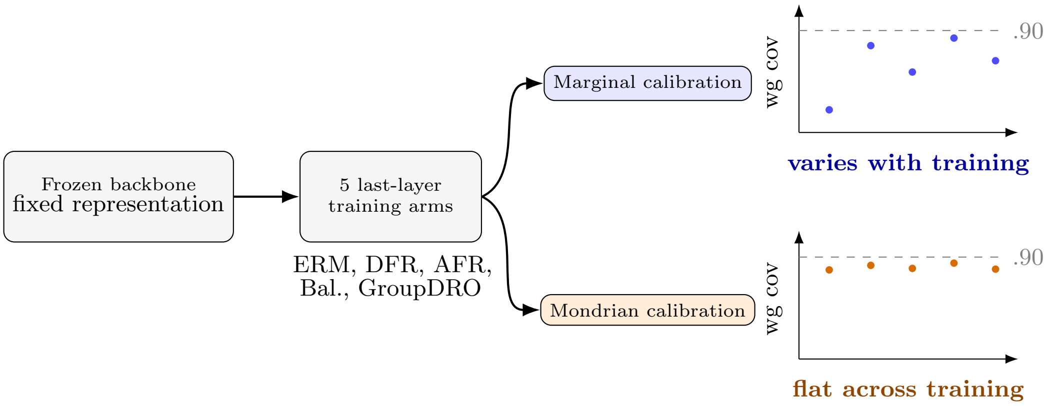

Fig. 1 Overview of the dissociation. The representation—a frozen backbone with five last-layer training arms—is held fixed; only the calibration policy changes. Under *marginal* calibration, worst-group coverage varies sharply with the training method and ERM collapses; under *Mondrian* (group-conditional) calibration it is flat across training methods and near target. Worst-group coverage is a property of the calibration mechanism, not of the representation.

last-layer arms are drawn from this zoo so that our worst-group accuracies are comparable to published values; our object of study is what these interventions do to the *conformal* side, which this literature does not measure.

Group-conditional and fair conformal prediction.

Mondrian conformal prediction (Vovk et al. 2005) computes group-conditional thresholds; conditional guarantees beyond disjoint groups are given by Gibbs et al. (2024) and Jung et al. (2023), and Bairaktari et al. (2025) extend group-conditional coverage to overlapping and fractional memberships. On fairness, Cresswell et al. (2025) show that equalizing coverage can increase disparate impact and recommend equalizing set sizes; Zhou and Sesia (2024) adaptively equalize coverage while controlling efficiency; and Gao et al. (2026) prove that equalized coverage and equalized set size cannot in general hold simultaneously, recasting the calibration policy as a choice of *where* cross-group heterogeneity manifests. Our work is the empirical complement on the axis these papers hold fixed: we vary the *training* intervention (hence representation and score distribution) and ask whether the conserved heterogeneity shrinks at the source. It does not; the calibration policy remains decisive.

Conformal prediction under shift and for robust models.

Tibshirani et al. (2019) establish validity under covariate shift via likelihood reweighting; multi-distribution robust conformal prediction (Yang and Jin 2026) targets uniform worst-case coverage as a group-DRO analogue *inside the calibration procedure*, holding the model fixed—consistent with our finding that the calibration mechanism is the operative lever. No prior work crosses the training-intervention zoo with calibration policies under a correlation-strength sweep, nor controls for the accuracy confound we isolate.

Uncertainty, concepts, and the opposite direction.

Some recent work uses uncertainty to *improve* group robustness—e.g. evidential alignment that mitigates spurious bias without group labels (Ye et al. 2025)—the reverse of our direction. We treat conformal prediction over learned concept or concept-bottleneck spaces (Zhang et al. 2025; Stammer et al. 2024; Turan et al. 2026) as an interpretability layer, orthogonal to our thesis. Selective classification under group shift (Jones et al. 2020) found abstention can leave worst-group accuracy flat; our set-valued analyses extend this lineage to conformal prediction.

Group recovery, conditional coverage, and its limits.

Whether the protected group can be recovered is a separate question: biased ERM representations encode the spurious attribute strongly (the principle behind JTT and CNC), which our probe confirms (AUROC $\in [0.95, 0.99]$). This connects to conformal prediction under noisy class or group labels (Einbinder et al. 2024; Sesia et al. 2024), which corrects coverage but assumes the group is given; here the group is not noisy but *recoverable*, which removes the auditability barrier and, via predicted groups, the deployment barrier (Section 6.7). Finally, exact feature-conditional coverage is impossible distribution-free in finite samples (Barber et al. 2019), which is why group-*conditional* (Mondrian) coverage is the operative target; online and multivalid variants (Gibbs et al. 2024; Jung et al. 2023) tighten it, but our contribution holds the calibration family fixed and varies the training intervention to show the representation does not substitute for the mechanism.

3 Problem Setup and Notation

We observe triples (x, y, a) with image x, label $y \in \mathcal{Y}$, and binary spurious attribute $a \in \{0, 1\}$ (background/*place* for Waterbirds; *Male* for CelebA). Groups are $g = (y, a) \in \mathcal{G}$, $|\mathcal{G}| = 2|\mathcal{Y}|$. A frozen backbone ϕ maps x to features $\phi(x) \in \mathbb{R}^d$; a last-layer head f_θ trained by intervention m maps $\phi(x)$ to class scores. The *correlation strength* ρ is the evaluation probability that a aligns with y in the majority pattern; we calibrate at $\rho_{\text{cal}} = 0.95$ and evaluate over $\rho_{\text{test}} \in \{0.95, 0.9, 0.8, 0.7, 0.6, 0.5\}$, holding the calibration distribution fixed.

Conformal sets and calibration policies.

Given a nonconformity score $s(x, y)$ and a calibration set $\mathcal{D}_{\text{cal}}$, split conformal prediction computes $\hat{q} = \text{Quantile}_{1-\alpha}\big(\{s(x_i, y_i)\}_{i \in \mathcal{D}_{\text{cal}}}\big)$ and returns $C(x) = \{y \in \mathcal{Y} : s(x, y) \le \hat{q}\}$. We use three scores: THR ($s_{\text{THR}}(x, y) = 1 - \hat{p}(y \mid x)$); APS (Romano et al. 2020), the cumulative softmax mass up to y; and RAPS (Angelopoulos et al. 2021), APS with a tail regularizer. We compare three *calibration policies*: **marginal split** (one global $\hat{q}$); **Mondrian** (a separate $\hat{q}_g$ per group); and **shift-robust** (a reweighted/worst-case threshold over a total-variation ball around the calibration distribution; Appendix B). Target $1 - \alpha = 0.90$ throughout; the full configuration is in Appendix G.

Reliability quantities.

For group g, $\mathrm{cov}_g = \Pr[y \in C(x) \mid g]$ and $|C|_g$ is its mean set size. We report worst-group coverage $\min_g \mathrm{cov}_g$, the *coverage gap*

$$\Delta_{\mathrm{cov}} = \max_g \mathrm{cov}_g - \min_g \mathrm{cov}_g, \tag{1}$$

mean set size, and set-size disparity $\max_g |C|_g - \min_g |C|_g$.

Cross-group conformity-score divergence (the "burden").

Following the conserved-quantity view of Gao et al. (2026), we summarize cross-group score heterogeneity by the divergence between the worst group's conformity-score distribution P_{wg} and that of the remaining groups $P_{\neg\mathrm{wg}}$,

$$D = W_1(P_{\mathrm{wg}}, P_{\neg\mathrm{wg}}) \quad \text{(Wasserstein-1)}, \qquad D_{\mathrm{KS}} = \sup_t \big|F_{\mathrm{wg}}(t) - F_{\neg\mathrm{wg}}(t)\big|, \tag{2}$$

computed per score on a held-out split; D is reported per score (scales differ).

A dissociation principle.

The empirical study is organized around the following elementary fact, which separates the by-design content of group-conditional calibration from the substantive question we test.

Proposition 1 (Calibration–representation dissociation) *Fix a score s and target $1-\alpha$, and assume within-group exchangeability of calibration and test scores.* (i) *Group-conditional (Mondrian) split conformal, with per-group threshold $\hat{q}_g$ the $\lceil (1-\alpha)(n_g+1)\rceil$-th smallest calibration score in group g, satisfies $\Pr\big(Y \in C(X) \mid G = g\big) \geq 1-\alpha$ for every group g, with no dependence on s beyond exchangeability.* (ii) *Marginal split conformal, with a single global threshold $\hat{q}$ set to the pooled $(1-\alpha)$-quantile, has per-group coverage $\Pr\big(Y \in C(X) \mid G = g\big) = F_g(\hat{q})$, where F_g is the group-g nonconformity-score CDF; the worst-group shortfall $(1-\alpha) - \min_g F_g(\hat{q})$ increases as the worst group's score distribution P_{wg} shifts away from the pool, i.e. with the cross-group divergence D of (2). Hence the representation (through s, and through any training intervention that changes s) controls worst-group coverage* only *under marginal calibration; under Mondrian it does not.*

Proof sketch (i) is the standard finite-sample split-conformal guarantee applied within each group under within-group exchangeability. For (ii), $\hat{q}$ is fixed by the pooled calibration scores; by definition the group-g coverage is $F_g(\hat{q})$. The worst group is the one whose scores are most stochastically enlarged (highest nonconformity); as P_{wg} is pushed further above the pool—which W_1 and the KS statistic of (2) quantify—$F_{\mathrm{wg}}(\hat{q})$ falls further below $1-\alpha$. □

Remark (what is and is not tested). Part (i) is true by construction; our contribution is therefore *not* that Mondrian covers groups, but the *quantitative* claim of part (ii)—that under spurious correlation the relevant divergence D, and hence the marginal-calibration shortfall, is governed by base accuracy and is *not* reduced by last-layer training beyond its accuracy effect (Sections 6.1, 6.3). Proposition 1 also predicts the observed sub-target behavior: Mondrian's $\geq 1-\alpha$ guarantee holds in the exchangeable limit, so the empirical 0.85–0.89 reflects finite-sample variance at small worst-group calibration counts together with the mild exchangeability perturbation induced by evaluating at ρ_{test} away from ρ_{cal}—a sub-target *baseline*, not a shift collapse (Section 6.5).

Group recoverability.

To test whether the deployed representation hides the protected group, we measure the test AUROC of a probe $h : \phi(x) \mapsto a$ trained in-domain to recover the spurious attribute.

4 Confound-Controlled Evaluation Protocol

Accuracy-matched divergence.

D in (2) and Δ_{cov} in (1) are mechanically entangled with base accuracy: a more accurate model produces sharper, more group-aligned score distributions, so a naive "robust training lowers D" partly restates "robust training raises accuracy." For each method m and the ERM reference we collect per-seed pairs $\left(\text{acc}_m^{(s)}, D_m^{(s)}\right)$ and read the divergence at a common accuracy $a^\star$ in the *overlapping* accuracy support:

$$\Delta(a^\star) = D_{\text{ERM}}(a^\star) - D_m(a^\star), \qquad \Delta(a^\star) > 0 \Rightarrow m \text{ lowers burden at matched accuracy,} \tag{3}$$

with a 95% bootstrap CI over seeds. When the accuracy supports of m and ERM do *not* overlap, $\Delta(a^\star)$ is **undefined** (`unmatched`); we never substitute the raw D, which we report separately, labeled uncontrolled.

Pre-registered hypotheses.

Hypothesis 1 (Coverage is a calibration property) *Under Mondrian, worst-group coverage is near target for every training method including ERM, while under marginal split it falls below target by an amount that depends on the training method—so the calibration policy, not the representation, controls worst-group coverage.*

Hypothesis 2 (Training buys efficiency) *Holding Mondrian fixed, worst-group set size differs by training method and tracks base accuracy.*

Hypothesis 3 (Transfer) *Robust training lowers the accuracy-matched divergence relative to ERM (CI of (3) excludes 0 on $\geq 2/3$ scores at ρ_{cal}).*

Hypothesis 4 (Ranking inversion) *The training method best for worst-group accuracy differs from the one best for worst-group conformal burden, with a CI-separated Δ_{cov} gap.*

Hypothesis 5 (Shift survival) *Any reduction or coverage level holds across the ρ_{test} sweep.*

Table 1 Cached split sizes (sample-level) and structure. Both datasets are binary with four spurious groups; the evaluation distribution is re-composited to a target correlation strength ρ (see below).

Dataset	$\lvert\mathcal{Y}\rvert$	$\lvert\mathcal{G}\rvert$	train ($\times d$)	eval ($\times d$)
Waterbirds	2	4	4795	5245
CelebA	2	4	162,770	29,872

Hypothesis 6 (No auditability tension) *The spurious attribute remains recoverable from the representation on which worst-group coverage is obtained, so coverage and post-hoc group auditing coexist.*

All may return negative; we report whatever the data show.

Discipline.

Each (backbone, dataset, method, seed) arm must reach a sane worst-group accuracy for its method or it is excluded with a logged reason; all probes are standardized (StandardScaler $\rightarrow$ logistic regression) and trained in-domain, with no test-label leakage. We use ≥ 3 training seeds $\times \geq 10$ calibration splits and report training-seed and calibration-split variance separately.

5 Experimental Setup

Datasets and backbones.

Waterbirds and CelebA (blond-hair classification, *Male* spurious). Two frozen backbones give the breadth axis: an ERM-trained ResNet-50 (d=2048, the canonical setup) and CLIP ViT-B/32 (d=512). Features are cached once. Both tasks are binary ($|\mathcal{Y}| = 2$) with four groups $g = (y, a)$; cached split sizes are in Table 1 and exact per-group counts are released with the artifacts.

Correlation-strength construction.

For a target ρ we re-weight / re-composite the evaluation rows so that the spurious attribute a aligns with the majority label pattern with probability ρ, holding the marginal class balance fixed; the realized ρ is logged per draw. Calibration uses $\rho_{\text{cal}} = 0.95$; final-test coverage is read at each $\rho_{\text{test}} \in \{0.95, 0.9, 0.8, 0.7, 0.6, 0.5\}$.

Calibration policies.

Beyond the definitions above, Mondrian falls back to the pooled quantile when a group's calibration count is below a minimum, and shift-robust uses a total-variation ball of radius ϵ (the higher quantile maintaining coverage for any reweighting within the ball), trading efficiency for stability under shift.

Uncertainty.

All point estimates are means over ≥ 3 training seeds $\times \geq 10$ calibration splits; 95% confidence intervals are nonparametric bootstrap intervals ($B = 1000$) over the (seed, split) cells, and we report training-seed and calibration-split standard deviations separately so model and calibration variance are not conflated.

Training interventions (last-layer arms).

On frozen features we fit ERM, DFR, AFR, balanced subsampling, and last-layer GroupDRO (minutes each on cached features). A full-network GroupDRO fine-tune is omitted as unnecessary for any conclusion and to avoid an underpowered single heavy arm.

Quality-gate outcomes.

All arms pass except AFR on CelebA, which collapses (worst-group accuracy 0.42 on ResNet-50, 0.013 on CLIP) and is excluded; last-layer GroupDRO seed 1 on CelebA/ResNet-50 (0.676) is excluded. These exclusions do not affect the dissociation (H1): cross-training spreads are computed over the retained arms and remain ≤ 0.024 under Mondrian including ERM, so dropping AFR (which fails its own robustness gate) cannot inflate or create the effect. Worst-group accuracy confirms the gate: ERM is far below parity on the balanced distribution (0.35–0.51, strong shortcut learning) while the robust arms raise it to 0.64–0.91 (DFR best on Waterbirds; balanced and GroupDRO-LL best on CelebA); exclusions are in Appendix F and full per-arm accuracies in the released records.

6 Results

6.1 Coverage Is a Calibration Property, Not a Representation One (H1)

Our central experiment holds the model fixed and varies only the calibration policy. Table 2 reports worst-group coverage (APS, $\rho_{\text{cal}} = 0.95$) under marginal split, Mondrian, and shift-robust calibration, for every training method and cell; Figure 2 visualizes it.

The dissociation is uniform. Under *marginal* calibration, worst-group coverage is governed by the training method: it ranges from 0.51 (ERM, CLIP/Waterbirds) to 0.87 (balanced, CLIP/Waterbirds), a spread of up to 0.36, with ERM always the worst. Under *Mondrian*, worst-group coverage is essentially flat across all training methods in every cell—cross-training spreads of 0.008 (ResNet/Waterbirds), 0.024 (CLIP/Waterbirds), 0.003 (ResNet/CelebA), and 0.004 (CLIP/CelebA)—and the worst representation is lifted to match the best. The cleanest single comparison: the *same* ERM model, re-calibrated from marginal to Mondrian, jumps from 0.564 to 0.870 (ResNet/Waterbirds) and from 0.509 to 0.854 (CLIP/Waterbirds), reaching the same worst-group coverage as DFR. On CelebA, where ERM's marginal coverage is less catastrophic (0.82), the Mondrian lift is correspondingly smaller (+0.06) but identical in direction. Mondrian's worst-group coverage settles at 0.85–0.89, slightly below

Table 2 H1: worst-group coverage (APS, $\rho_{\text{cal}} = 0.95$) by calibration policy and training method. Under marginal split, coverage depends strongly on training (ERM worst); under Mondrian it is flat across training in every cell and the same ERM model is lifted to match the robust arms. RAPS/THR in Appendix A.

Backbone/Dataset	Method	marginal split	Mondrian	shift-robust
ResNet-50 / Waterbirds	ERM	**0.564**	0.870	0.798
	DFR	0.854	0.868	0.948
	AFR	0.809	0.865	0.941
	Bal. subsample	0.711	0.873	0.888
	GroupDRO-LL	0.707	0.873	0.865
	spread	*0.290*	*0.008*	
CLIP / Waterbirds	ERM	**0.509**	0.854	0.653
	DFR	0.852	0.865	0.946
	AFR	0.805	0.857	0.923
	Bal. subsample	0.867	0.864	0.958
	GroupDRO-LL	0.693	0.841	0.844
	spread	*0.358*	*0.024*	
ResNet-50 / CelebA	ERM	**0.821**	0.881	0.902
	DFR	0.877	0.881	0.922
	Bal. subsample	0.877	0.878	0.928
	GroupDRO-LL	0.892	0.878	0.939
	spread	*0.071*	*0.003*	
CLIP / CelebA	ERM	**0.822**	0.886	0.921
	DFR	0.880	0.882	0.918
	Bal. subsample	0.879	0.885	0.921
	GroupDRO-LL	0.882	0.884	0.921
	spread	*0.060*	*0.004*	

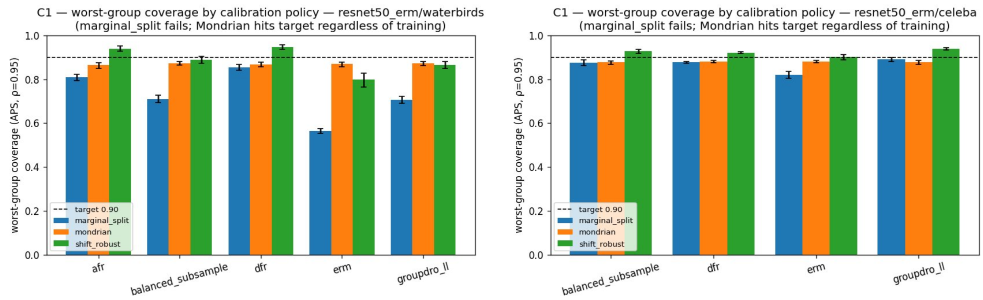

Fig. 2 H1. Worst-group coverage by calibration policy, grouped by training method, for two representative cells (all four in Table 2): *left*, ResNet-50/Waterbirds (large ERM lift); *right*, CLIP/CelebA (smallest, near target). Marginal split (blue) varies with training and fails for ERM; Mondrian (orange) is flat across training methods in every cell. The representation does not deliver worst-group coverage; the calibration mechanism does.

the 0.90 target—a finite-sample residual we report separately (Section 6.5) that does not bear on the dissociation. **Worst-group coverage is delivered by the calibration mechanism; the representation does not provide it.** RAPS and THR reproduce the dissociation, THR with slightly larger residual variance (Appendix A).

Table 3 H2: worst-group set size under Mondrian (APS, $\rho_{\text{cal}} = 0.95$) vs. worst-group accuracy. "corr" is Pearson correlation of (worst-group accuracy, worst-group set size) across methods. Negative = more accurate yields smaller sets.

Backbone/Dataset	Method	wg set size	wg acc
ResNet-50 / Waterbirds (corr +0.04)	ERM	1.138	0.510
	DFR	1.133	0.812
	AFR	1.218	0.686
	Bal. subsample	1.159	0.637
	GroupDRO-LL	1.101	0.656
CLIP / Waterbirds (corr −0.85)	ERM	1.319	0.485
	DFR	1.135	0.838
	AFR	1.300	0.688
	Bal. subsample	1.175	0.796
	GroupDRO-LL	1.213	0.647
ResNet-50 / CelebA (corr −0.97)	ERM	1.351	0.416
	DFR	1.092	0.854
	Bal. subsample	1.197	0.757
	GroupDRO-LL	1.091	0.808
CLIP / CelebA (corr −1.00)	ERM	1.163	0.352
	DFR	1.047	0.879
	Bal. subsample	1.056	0.890
	GroupDRO-LL	1.041	0.905

6.2 Training Buys Efficiency, Not Coverage (H2)

Given that Mondrian equalizes worst-group *coverage* across training methods, where does training help? Table 3 reports the worst-group set size under Mondrian against base accuracy. In three of four cells the worst-group set size tracks base accuracy (Pearson −0.85 on CLIP/Waterbirds, −0.97 on ResNet/CelebA, −1.00 on CLIP/CelebA): more accurate training yields smaller (more efficient) worst-group sets at the *same* coverage. ResNet/Waterbirds is the exception (+0.04, no relationship). Because $|\mathcal{Y}| = 2$ the absolute efficiency range is narrow, but the direction is consistent. Training and accuracy therefore buy *efficiency*; the calibration mechanism buys coverage.

6.3 "Transfer" Is Mostly an Accuracy Effect (H3)

The dissociation above predicts that any apparent training-side reduction of the conformal *burden* should be largely an accuracy effect. Table 4 confirms it. The *raw* divergence collapses uniformly—ERM's APS divergence of 0.19–0.22 falls to 0.01–0.12 for the robust arms—but base accuracy also moves sharply, so the raw number is confounded. Once *matched*, the picture fragments: on Waterbirds the methods whose accuracy overlaps ERM (GroupDRO-LL on both backbones; balanced on ResNet-50) show a significant but *modest* residual reduction (e.g. GroupDRO-LL on CLIP/Waterbirds $\Delta_{\text{APS}} = +0.100\,[+0.085, +0.117]$), while on *CelebA every arm is*

Table 4 H3: accuracy-matched divergence $\Delta(a^\star)$ (APS, $\rho_{\text{cal}} = 0.95$; > 0 means robust lowers burden at matched accuracy) and uncontrolled raw divergence with base top-1. `unm.` = unmatched. Full per-score table in Appendix C.

Backbone/Dataset	Method	$\Delta_{\text{APS}}(a^\star)$ [CI]	raw D_{APS}	base top-1
ResNet-50 / Waterbirds	ERM (ref)	—	0.190	0.940
	DFR	`unm.`	0.038	0.864
	Bal. subsample	$+0.059\,[+0.030, +0.085]$	0.104	0.918
	GroupDRO-LL	$+0.078\,[+0.061, +0.094]$	0.109	0.939
CLIP / Waterbirds	ERM (ref)	—	0.216	0.932
	DFR	`unm.`	0.037	0.870
	Bal. subsample	`unm.`	0.034	0.836
	GroupDRO-LL	$+0.100\,[+0.085, +0.117]$	0.118	0.928
ResNet-50 / CelebA	ERM (ref)	—	0.190	0.699
	DFR	`unm.`	0.027	0.862
	Bal. subsample	`unm.`	0.032	0.809
	GroupDRO-LL	`unm.`	0.013	0.859
CLIP / CelebA	ERM (ref)	—	0.197	0.677
	DFR	`unm.`	0.030	0.902
	Bal. subsample	`unm.`	0.025	0.904
	GroupDRO-LL	`unm.`	0.023	0.912

unmatched—robust training raises base accuracy to 0.81–0.91 versus ERM's 0.68–0.70, so no common accuracy level exists and $\Delta(a^\star)$ is undefined exactly where robustness is strongest (CelebA, Table 4). Uncontrolled comparisons thus overstate the conformal-specific benefit by roughly an order of magnitude ($0.19 \to 0.03$ raw vs. a $\sim$0.06–0.10 matched residual where it is even defined).

6.4 Worst-Group Accuracy Is a Good Proxy for Conformal Burden (H4)

Table 5 reports worst-group accuracy and conformal burden Δ_{cov} with the inversion verdict. An inversion (accuracy-best $\neq$ burden-best, CI-separated) occurs in only 2 of 4 cells, with no consistent pattern—on CLIP for Waterbirds but on ResNet-50 for CelebA—and is marginal where it appears on CelebA (Δ_{cov} gap $+0.010\,[+0.0005, +0.019]$, both methods near the coverage ceiling). Across cells the two rankings are strongly concordant: worst-group accuracy is a reasonable proxy for worst-group conformal burden.

6.5 Stable but Sub-Target under Shift; No Relocation (H5)

We sweep ρ_{test} with calibration fixed at $\rho_{\text{cal}} = 0.95$ (Figure 3; full tables in Appendix E). Three observations hold across cells. (i) *Stability*: worst-group coverage varies by < 0.05 over the whole sweep for every method. (ii) *Sub-target*: under Mondrian no method reaches 0.90; the stable level is 0.85–0.89, consistent with group-conditional calibration relocating but not closing the worst-group gap. Whether this level clears a strict "≥ 0.88" validity bar across the sweep is therefore baseline-dependent: it does on CelebA (level ≈ 0.88) and does not on Waterbirds (level ≈ 0.85), but the curve is flat in both—this is a sub-target baseline, not a shift collapse. (iii)

Table 5 H4: worst-group accuracy and burden Δ_{cov} (lower better, 95% CI) for the accuracy-best and burden-best methods per cell. Inversion is real in 2/4 cells and marginal on CelebA. Full five-method table in Appendix D.

Backbone/Dataset	Method (acc / burden-best)	wg acc	Δ_{cov} [CI]	inversion
ResNet-50 / Waterbirds	DFR (both) GroupDRO-LL	0.812 0.656	0.047 [0.040,0.055] 0.246 [0.229,0.263]	no
CLIP / Waterbirds	DFR (acc) Bal. subsample (burden)	0.838 0.796	0.053 [0.046,0.061] 0.035 [0.029,0.042]	**real** Δ=+0.018
ResNet-50 / CelebA	DFR (acc) GroupDRO-LL (burden)	0.854 0.808	0.022 [0.019,0.025] 0.012 [0.004,0.021]	**real** (marg.) Δ=+0.010
CLIP / CelebA	GroupDRO-LL (both) Bal. subsample	0.905 0.890	0.013 [0.010,0.015] 0.018 [0.015,0.020]	no

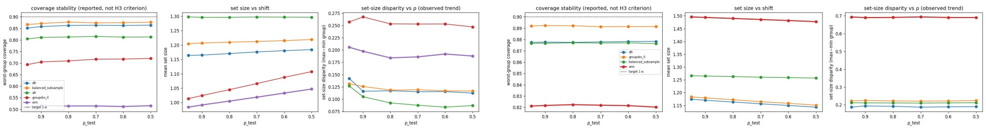

Fig. 3 H5 under correlation-strength shift (ρ_{test} swept with calibration fixed at $\rho_{\text{cal}} = 0.95$). Within each panel: worst-group coverage (stable, sub-target), mean set size, and set-size disparity (mostly easing as $\rho \to 0.5$). Worst-group coverage stays flat and below the dashed 0.90 target across the sweep; there is no relocation of the conformal burden.

No relocation: the H1 matched-divergence reduction becomes `undefined` across the sweep, and set-size disparity *eases* as $\rho \to 0.5$ in most cells rather than inflating (only DFR and GroupDRO-LL on CelebA/ResNet-50 show mild growth). The "relocate, not remove" phenomenon of Gao et al. (2026) is therefore not observed as a shift effect: the burden simply tracks correlation strength and is maximal at the calibration point.

The sub-target level is finite-sample, not a failure of the dissociation: Mondrian worst-group coverage is 2–4× more stable across calibration splits than marginal (SD 0.011–0.036 vs. 0.030–0.138), and the shortfall co-scales with this SD—larger on Waterbirds (0.030–0.044) than on CelebA (0.016–0.020). This co-scaling is the signature of finite-sample worst-group quantile noise: a stable, near-target baseline whose gap shrinks as worst-group calibration data grows.

6.6 No Coverage–Auditability Tension (H6)

Finally we test whether the representation that yields worst-group coverage hides the protected group. A standardized probe recovers the spurious attribute at AUROC = $0.947\,[0.945, 0.957]$ (ResNet/Waterbirds) and $0.977\,[0.975, 0.982]$ (CLIP/Waterbirds) while Mondrian simultaneously achieves worst-group coverage $0.870\,[0.802, 0.903]$ and $0.854\,[0.791, 0.881]$: coverage and auditability *coexist*. Projecting out the top-k spurious-predictive directions (Figure 4) leaves coverage flat on ResNet-50 and raises it by at most 0.02 on CLIP—still sub-target—as AUROC falls toward chance. Forcing invariance does not unlock coverage; group-conditional calibration already delivers it

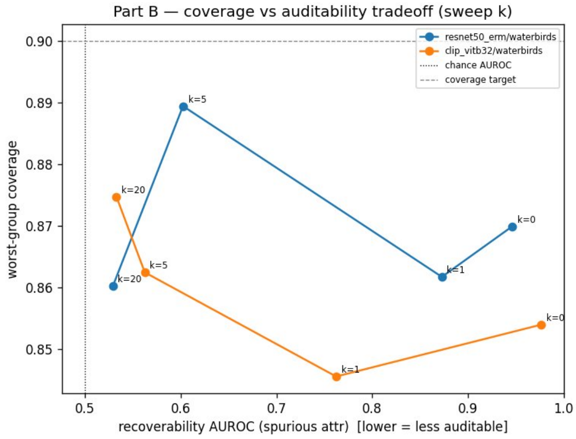

Fig. 4 Worst-group coverage vs. spurious-attribute recoverability as the top-k spurious directions are projected out ($k \in \{0, 1, 5, 20\}$). Coverage is flat (ResNet-50) or rises only $\sim$0.02 and stays sub-target (CLIP) as recoverability falls toward chance.

from a fully recoverable representation, so there is no coverage–auditability tension, consistent with H1.

6.7 Group-Conditional Calibration Is Deployable with Predicted Groups

Mondrian uses group labels for per-group thresholds, raising a deployment objection: are ground-truth groups required? Because the spurious attribute is highly recoverable (Section 6.6), they are not. We re-run Mondrian with probe-predicted groups in three conditions: (a) true groups; (b) predicted groups at test only (thresholds from true-group calibration); and (c) predicted groups at *both* calibration and test—the fully deployable setting in which the attribute is never observed. Coverage is always scored on the true groups.

Table 6 reports the deployable setting (c) against the oracle (a) for APS. The hit is small: 16 of 18 (method, cell) combinations clear a 0.02 bar—every method on the ResNet cells ($|\mathrm{cov}(a) - \mathrm{cov}(c)| \leq 0.012$) and all but one robust method on the CLIP cells (≤ 0.017). The two exceptions are instructive: ERM on CLIP/Waterbirds (gap $+0.051$), the most group-dependent model, where probe errors on the minority group hurt most and which one would not deploy for worst-group reliability anyway; and a marginal overshoot for GroupDRO on CLIP/CelebA ($+0.025$). THR is the least stable score (Appendix H); APS and RAPS are robust. This closes the loop with H6: the *same* recoverability that makes the representation auditable also makes group-conditional calibration deployable without observed group labels.

7 Discussion

The results compose into a single thesis. Worst-group *coverage* is delivered by the calibration policy: marginal calibration leaves it at the mercy of the representation

Table 6 Deployable Mondrian (APS, ρ_{cal}=ρ_{test}=0.95): worst-group coverage under ground-truth groups cov(a) vs. probe-predicted groups at calibration and test cov(c), with the gap and worst-group set-size cost. Deployable = $|\mathrm{cov}(a) - \mathrm{cov}(c)| \leq 0.02$. Probe AUROC in $[0.945, 0.999]$.

Backbone/Dataset	Method	cov(a)	cov(c)	gap	size cost	deploy.
ResNet-50 / Waterbirds	AFR	0.865	0.877	−0.012	+0.01	✓
	Bal. subsample	0.873	0.867	+0.006	−0.02	✓
	DFR	0.868	0.874	−0.007	+0.02	✓
	ERM	0.870	0.861	+0.009	+0.05	✓
	GroupDRO-LL	0.873	0.870	+0.003	+0.08	✓
ResNet-50 / CelebA	Bal. subsample	0.878	0.879	−0.001	+0.02	✓
	DFR	0.881	0.878	+0.003	−0.02	✓
	ERM	0.881	0.876	+0.005	−0.12	✓
	GroupDRO-LL	0.878	0.875	+0.002	−0.04	✓
CLIP / Waterbirds	AFR	0.857	0.855	+0.002	+0.06	✓
	Bal. subsample	0.864	0.873	−0.008	+0.01	✓
	DFR	0.865	0.869	−0.004	+0.02	✓
	ERM	0.854	0.803	+0.051	+0.20	—
	GroupDRO-LL	0.841	0.826	+0.015	+0.13	✓
CLIP / CelebA	Bal. subsample	0.885	0.880	+0.005	−0.01	✓
	DFR	0.882	0.865	+0.017	−0.06	✓
	ERM	0.886	0.887	−0.001	−0.02	✓
	GroupDRO-LL	0.884	0.859	+0.025	−0.05	—

(ERM collapses), while Mondrian makes it nearly invariant to training and lifts even ERM to match a robustly trained model (H1); training instead buys *efficiency* (H2). Everything else follows: the apparent training-side reduction of conformal burden is largely an accuracy effect, modest or unmeasurable once matched (H3); accuracy-best is almost always burden-best (H4); under shift coverage is stable but sub-target (H5); and the representation does not hide the group, so coverage and auditability coexist (Section 6.6). The lever is the calibration *mechanism*, not the score *representation*.

The recommendation is deflationary and useful: (a) report conformal-burden comparisons *at matched accuracy*, since uncontrolled comparisons overstate training-side benefits by an order of magnitude; (b) do not expect a training method chosen for worst-group accuracy to be suboptimal for conformal burden; and (c) obtain worst-group coverage from group-conditional calibration, which remains effective on ordinary, fully auditable representations and—because the group is recoverable—is deployable with probe-predicted groups (Section 6.7), so test-time group labels are not required.

8 Limitations

Our conclusions are empirical and scoped to two vision benchmarks, two frozen backbones, last-layer interventions, and binary spurious attributes; a full-network GroupDRO fine-tune and a third dataset would extend breadth, though the consistency of H1 across all four cells suggests additional data would refine rather than overturn it. The accuracy-matched comparison (H3) is undefined when a method's accuracy support does not overlap ERM's, which is itself informative but prevents a uniform statement. Because $|\mathcal{Y}| = 2$, set sizes lie in $[1, 2]$, compressing the efficiency axis (H2); multi-class tasks would give it more range. Mondrian's group-conditional coverage is, by construction, close to a per-group target, so H1's non-trivial content is the *magnitude* of marginal calibration's representation-dependent failure and the size of the ERM lift, which we quantify. Deployable Mondrian (Section 6.7) holds for APS/RAPS and robust arms but degrades for the most group-dependent model and the THR score, scoping that claim to robust models and rank-based scores; it also infers a protected attribute at test time, so it should be used only where such inference is permissible and auditable. Confidence intervals do not model cross-dataset variance.

9 Conclusion

Across a confound-controlled grid, the honest answers are: worst-group coverage is a calibration property, not a representation one; training buys efficiency, not coverage; the conformal "transfer" is mostly an accuracy effect; the inversion is rare; coverage is stable but sub-target under shift; and there is no auditability tension. We release the full $2 \times 2 \times 5 \times 3 \times 3$ grid, the shift sweep, and the accuracy-matched protocol so that future training-side conformal claims can be evaluated without the accuracy confound.

Data and code availability. Waterbirds and CelebA are public. All experiment records, the calibration ablation, predicted-group results, per-aggregate variance, the quality-gate log, and figure scripts are released: `grid_records.csv`, `calibration_ablation.csv`, `predicted_group_mondrian.csv`. The conformal layer re-runs in minutes on cached scores without GPU retraining.

Declarations

Funding. The authors received no specific funding for this work. **Competing interests.** The authors have no competing interests relevant to this article. **Ethics approval.** Not applicable; this study uses only public benchmark datasets and involves no human participants or animals. **Data and code availability.** See "Data and code availability" above. **Author contributions.** O.O. designed and ran the experiments and wrote the manuscript; N.Y. and Y.N. supervised the work and revised the manuscript. All authors approved the final version.

Appendix A Calibration-Ablation Details

Table 2 (main text) gives APS worst-group coverage under all three calibration policies for every cell and method; the coverage gap is $\Delta_{\text{cov}} = 0.90 - \text{cov}$ (negative = over-target). The dissociation is uniform across scores: RAPS and THR reproduce the marginal-dependent/Mondrian-flat pattern. The per-cell strict flag "Mondrian ≥ 0.88 for all *and* marginal below target for all" is met for APS and RAPS on CLIP/CelebA and fails elsewhere only because Mondrian's stable level sits at 0.85–0.88 (the sub-target residual of Section 6.5), not because the dissociation fails. Full per-(cell,score,method) coverage, Δ_{cov}, and CIs—including the shift-robust column, which over-covers (0.84–0.96) at an efficiency cost—are released in `calibration_ablation.csv`.

Appendix B Shift-Robust Calibration

The shift-robust policy generally over-covers the worst group relative to target (0.84–0.96) at the cost of larger sets, and is most useful when ρ_{test} is expected to move far from ρ_{cal}. It does not change the H1 conclusion: like Mondrian, it makes worst-group coverage far less representation-dependent than marginal calibration. Per-cell numbers are in Table 2 (shift-robust column) and the released CSV.

Appendix C Per-Score H3 (accuracy-matched) Tables

For each (cell, method) we report $\Delta(a^\star)$ for APS, RAPS, and THR with 95% CIs, the overlapping accuracy support, and the raw divergence with base accuracy. The GO bar (CI excludes 0 on $\geq 2/3$ scores) is met only by GroupDRO-LL (both Waterbirds backbones) and balanced subsampling (ResNet/Waterbirds); all CelebA arms are `unmatched` on all three scores. THR yields the largest matched reductions where defined (e.g. GroupDRO-LL on CLIP/Waterbirds $\Delta_{\text{THR}} = +0.159\,[+0.131, +0.185]$), consistent with THR's sensitivity to top-1 mass. Full numbers in the released records.

Appendix D Full Five-Method H4 Tables

Per cell, Δ_{cov} with 95% CIs for all retained methods, the worst-group accuracy ranking, the worst-group burden ranking, the point-estimate inversion, and the CI-separated inversion test. Summary of the inversion test: real on CLIP/Waterbirds ($\Delta\Delta_{\text{cov}} = +0.018\,[+0.008, +0.028]$) and ResNet/CelebA ($+0.010\,[+0.0005, +0.019]$); absent (CI includes 0) on ResNet/Waterbirds and CLIP/CelebA.

Appendix E Full H5 Shift Tables

Per (cell, method, score): the divergence-survival label across $\rho_{\text{test}} \in \{0.95, \ldots, 0.5\}$ (`survived` / `never_held` / `held_then_broke@`ρ / `undefined@`ρ), the set-size disparity at $\rho = 0.95$ and $\rho = 0.50$ with its observed trend (`eases`/`grows`/`flat`), and worst-group coverage mean, range over ρ, flatness (< 0.05), and sub-target flag. Across all cells: divergence survival is `undefined` (matched comparison loses overlap under shift); disparity trend is `eases` or `flat` except DFR/GroupDRO-LL on CelebA/ResNet-50 (`grows`); coverage range < 0.05 and sub-target for every method.

Appendix F Excluded and Flagged Arms

Hard-floor exclusions (worst-group accuracy below the per-method floor 0.75): AFR on CelebA (ResNet-50 0.42; CLIP 0.013, all seeds); last-layer GroupDRO seed 1 on CelebA/ResNet-50 (0.676). Soft flag (kept): DFR seed 2 on CelebA/ResNet-50 (0.847, just below the 0.85 reference). AFR's CelebA collapse is a known failure mode of automatic feature reweighting under this attribute and is not a pipeline error.

Appendix G Procedures, Hyperparameters, and Statistics

Features. Frozen ERM ResNet-50 (d=2048) and CLIP ViT-B/32 (d=512), extracted once and cached; all training is last-layer on cached features. **Heads.** ERM (cross-entropy); DFR (group-balanced last-layer retraining); AFR (automatic feature reweighting); balanced subsampling; last-layer GroupDRO. **Calibration.** marginal split (one global quantile); Mondrian (a separate quantile per group g, with a fall-back to the pooled quantile when a group's calibration count $n_g < 50$); shift-robust (worst-case threshold over a total-variation ball of radius $\epsilon = 0.1$ around the calibration distribution; results are insensitive over $\epsilon \in [0.05, 0.2]$, trading 1–5 points of over-coverage for stability); $1 - \alpha = 0.90$. **Scores.** THR, APS, RAPS (with standard tail regularization). **Shift.** $\rho_{\text{cal}} = 0.95$; $\rho_{\text{test}} \in \{0.95, 0.9, 0.8, 0.7, 0.6, 0.5\}$ via reweighted/composited evaluation distributions, with realized ρ logged. **Seeds/splits.** ≥ 3 training seeds $\times \geq 10$ calibration splits; 95% CIs by bootstrap; training-seed standard deviation and calibration-split standard deviation reported separately per aggregate. **Accuracy matching.** accuracy is overall (base) top-1 on the evaluation distribution; from the per-seed (acc, D) clouds of m and ERM we take $a^\star$ in the overlapping accuracy range and read each method's $D(a^\star)$ by linear interpolation of its cloud, bootstrapping over seeds; $\Delta(a^\star)$ is `unmatched` when the supports are disjoint. **Recoverability.** StandardScaler $\rightarrow$ logistic regression, in-domain train$\rightarrow$test; invariant-ization by projecting out the top-k spurious-predictive directions, $k \in \{0, 1, 5, 20\}$. **Predicted-group Mondrian.** the recoverability probe predicts the stratum $2y + \hat{a}$; thresholds and/or test assignment use $\hat{a}$; coverage is always scored on true groups. **Compute.** the full grid and ablation re-calibrate cached head posteriors; no run exceeds the budget. **Release.** `grid_records.csv`, `calibration_ablation.csv`, `predicted_group_mondrian.csv`, per-aggregate JSON (seed-std vs split-std), the quality-gate log, and all figure scripts are released.

Appendix H Predicted-Group Mondrian: RAPS and THR

Table 6 reports APS. RAPS reproduces it—both ResNet cells fully deployable, robust arms deployable on CLIP, ERM on CLIP/Waterbirds the exception. THR is the least stable score: under predicted groups several arms exceed the 0.02 bar in each cell (e.g. AFR and DFR on ResNet/Waterbirds via over-coverage, ERM and GroupDRO-LL on CLIP/Waterbirds), reflecting THR's sensitivity to small perturbations of the top-1 softmax mass when the stratum is mis-assigned. We therefore scope the deployable-Mondrian claim to APS/RAPS. Full per-(cell,score,method) coverage in all three conditions (true / predicted-test / predicted-both) with CIs are released in `predicted_group_mondrian.csv`.

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